11. Develop an Android application integrating SQLite with ListView or RecyclerView.

Project Name : SQLiteCRUDDemo
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">

   <EditText
       android:id="@+id/etName"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:hint="Enter Name" />

   <Button
       android:id="@+id/btnInsert"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:text="Insert" />
```

```
    <Button
        android:id="@+id/btnView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="View Data" />
    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />
</LinearLayout>
```

Create New Java Class in Package Folder

Name as: DBHelper

```
package com.example.sqlitecruddemo;


import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;


import java.util.ArrayList;


public class DBHelper extends SQLiteOpenHelper {


    public DBHelper(Context context) {
```

```
        super(context, "StudentDB", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE student(name TEXT)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        // Database upgrade code goes here
    }

    // Insert Data
    public void insertData(String name) {
        SQLiteDatabase db = this.getWritableDatabase();

        ContentValues values = new ContentValues();
        values.put("name", name);

        db.insert("student", null, values);
    }

    // Get All Data
    public ArrayList<String> getData() {
```

```
        ArrayList<String> list = new ArrayList<>();


        SQLiteDatabase db = this.getReadableDatabase();


        Cursor cursor = db.rawQuery("SELECT name FROM student", null);


        while (cursor.moveToNext()) {

            list.add(cursor.getString(0));

        }


        cursor.close();


        return list;

    }

}
```

MainActivity.java

```
package com.example.sqlitecruddemo;


import android.os.Bundle;

import android.widget.ArrayAdapter;

import android.widget.Button;

import android.widget.EditText;

import android.widget.ListView;

import android.widget.Toast;
```

```java
import androidx.appcompat.app.AppCompatActivity;


import java.util.ArrayList;


public class MainActivity extends AppCompatActivity {


   EditText etName;
   Button btnInsert, btnView;
   ListView listView;


   DBHelper dbHelper;


   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);


       setContentView(R.layout.activity_main);


       etName = findViewById(R.id.etName);
       btnInsert = findViewById(R.id.btnInsert);
       btnView = findViewById(R.id.btnView);
       listView = findViewById(R.id.listView);


       dbHelper = new DBHelper(this);
```

```
        // Insert Data
        btnInsert.setOnClickListener(v -> {

            String name = etName.getText().toString();

            dbHelper.insertData(name);

            Toast.makeText(
                    this,
                    "Data Inserted",
                    Toast.LENGTH_SHORT
            ).show();

            etName.setText("");
        });

        // View Data
        btnView.setOnClickListener(v -> {

            ArrayList<String> list = dbHelper.getData();

            ArrayAdapter<String> adapter = new ArrayAdapter<>(
                    this,
                    android.R.layout.simple_list_item_1,
                    list
            );
```

```
            listView.setAdapter(adapter);

        });

    }

}
```

